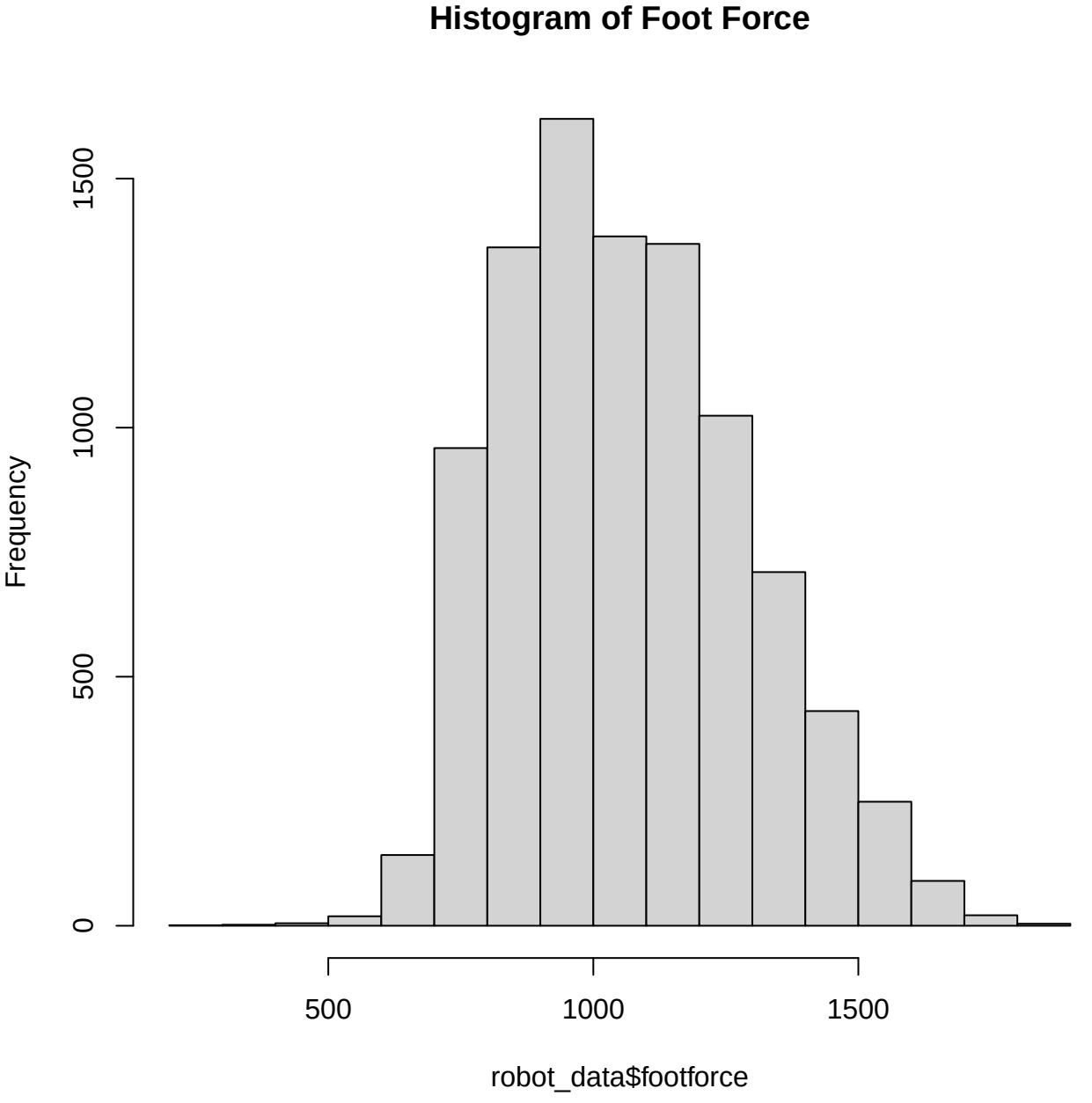
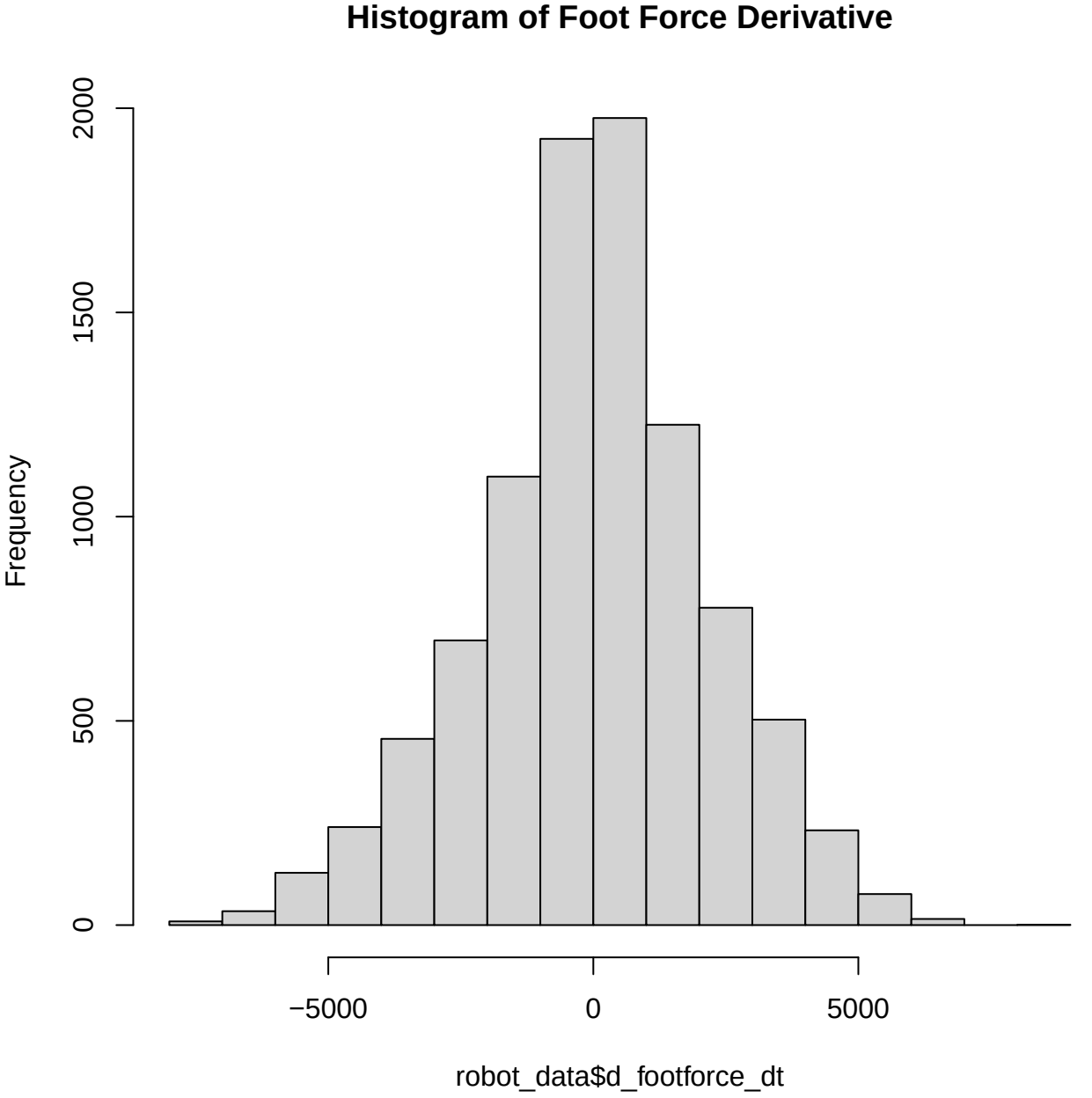
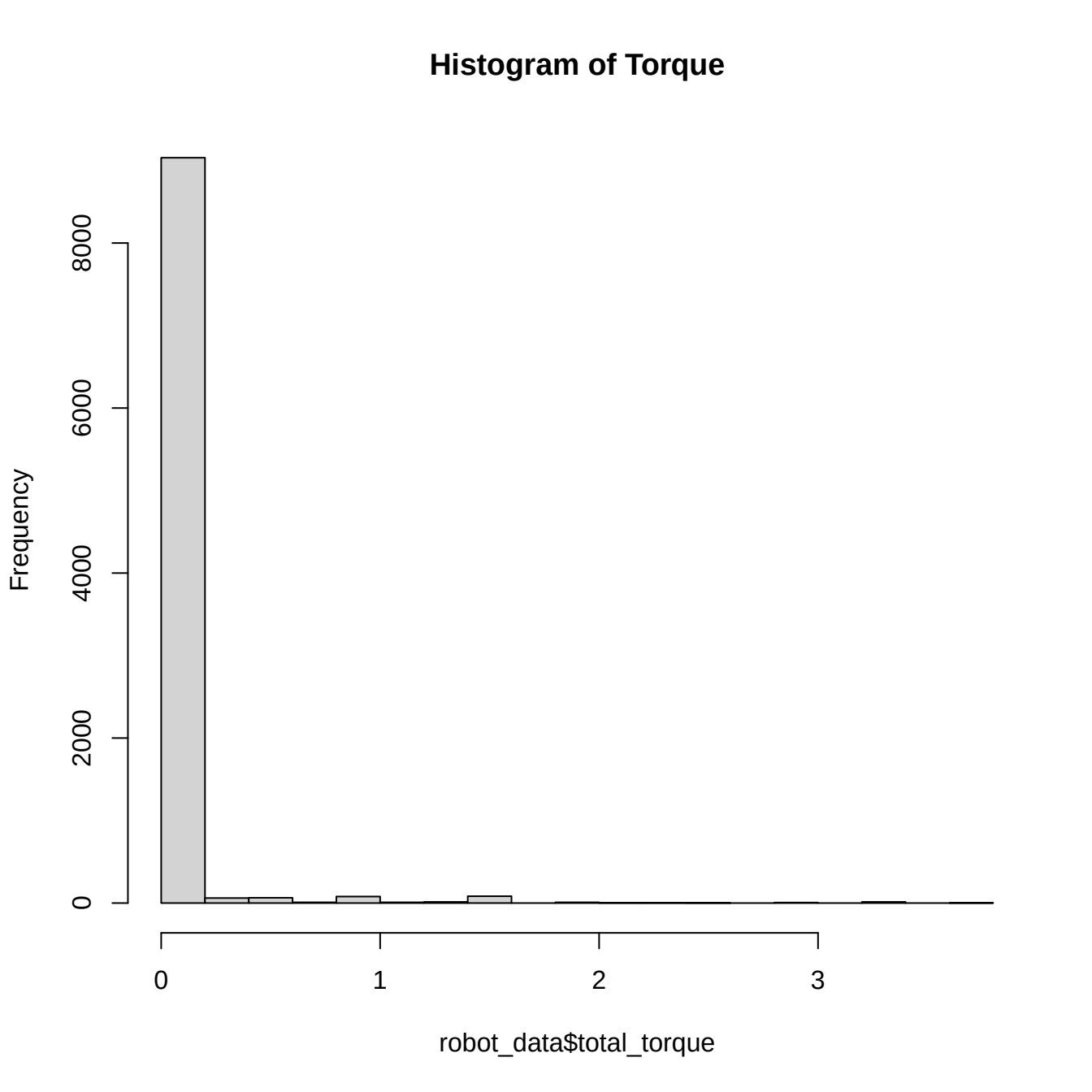
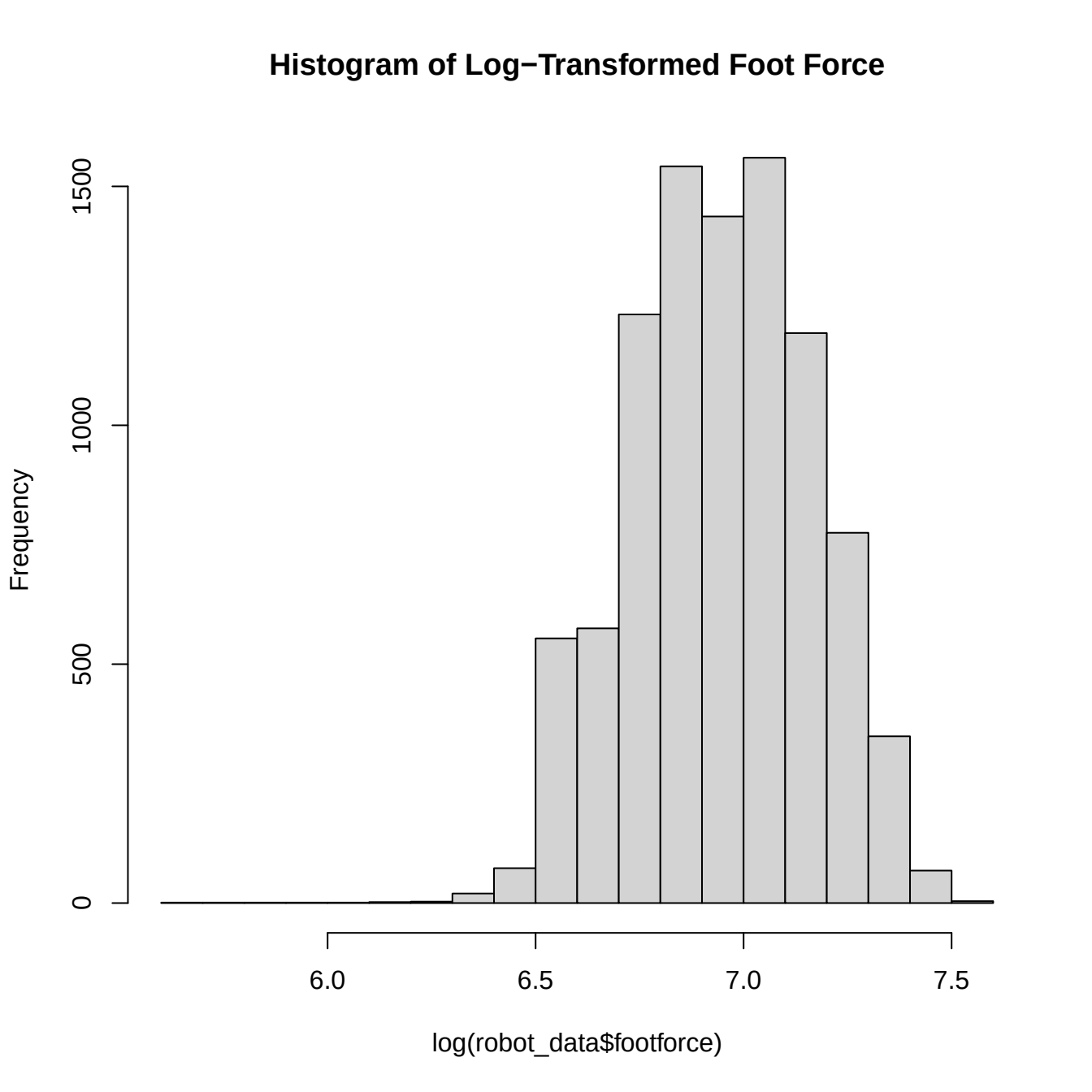
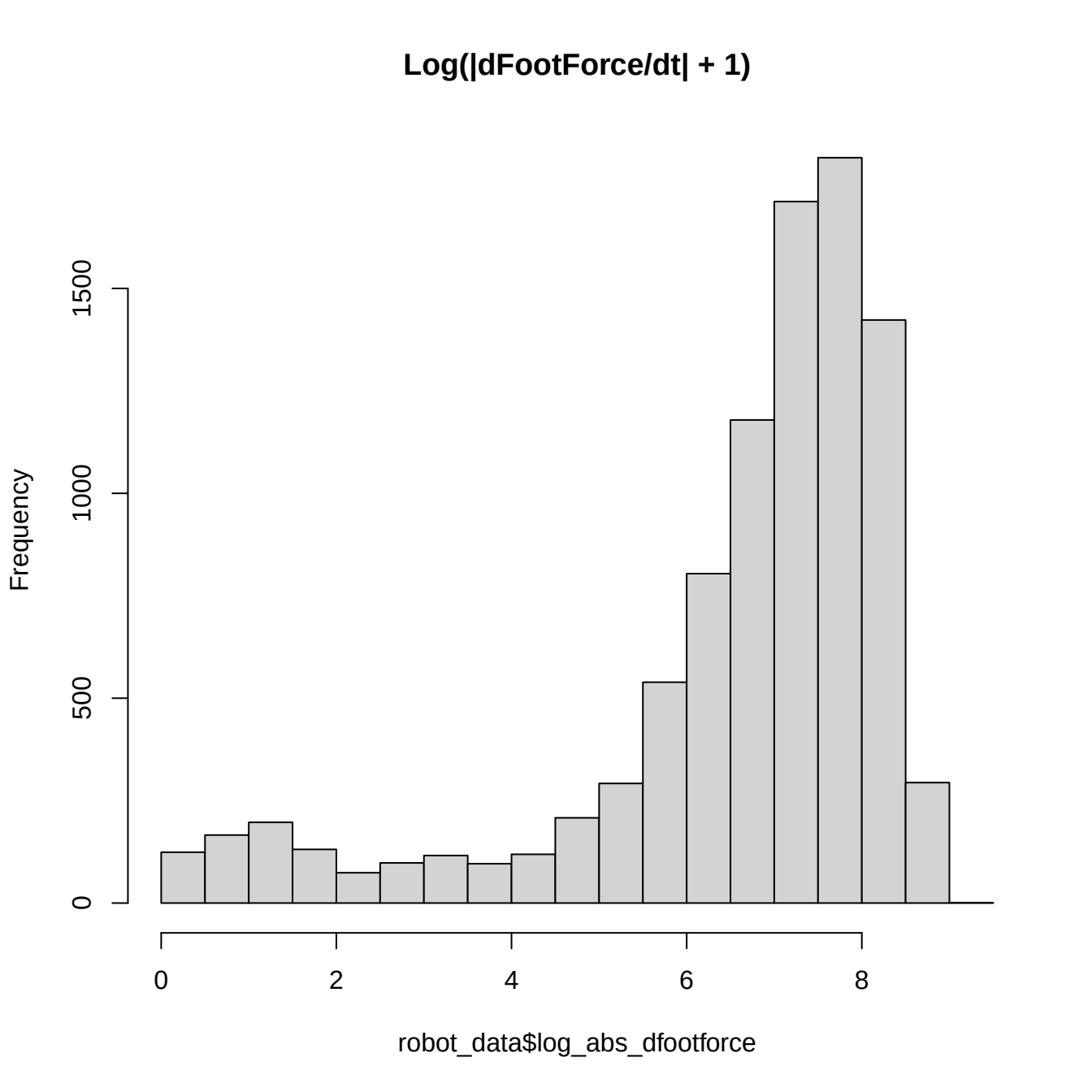
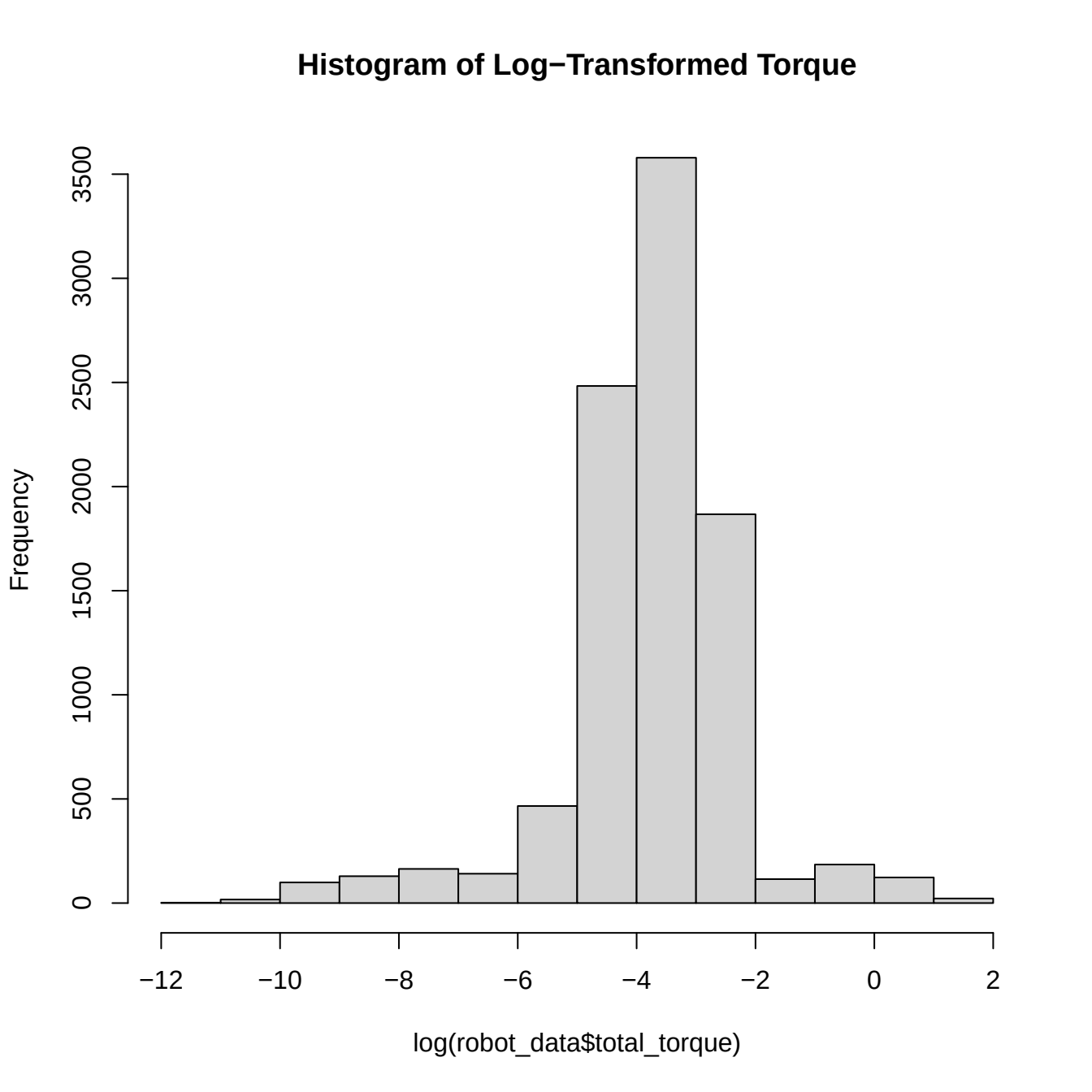

Histogram of Foot Force
Frequency
0
500
1000
1500
500
1000
1500
robot_data$footforce

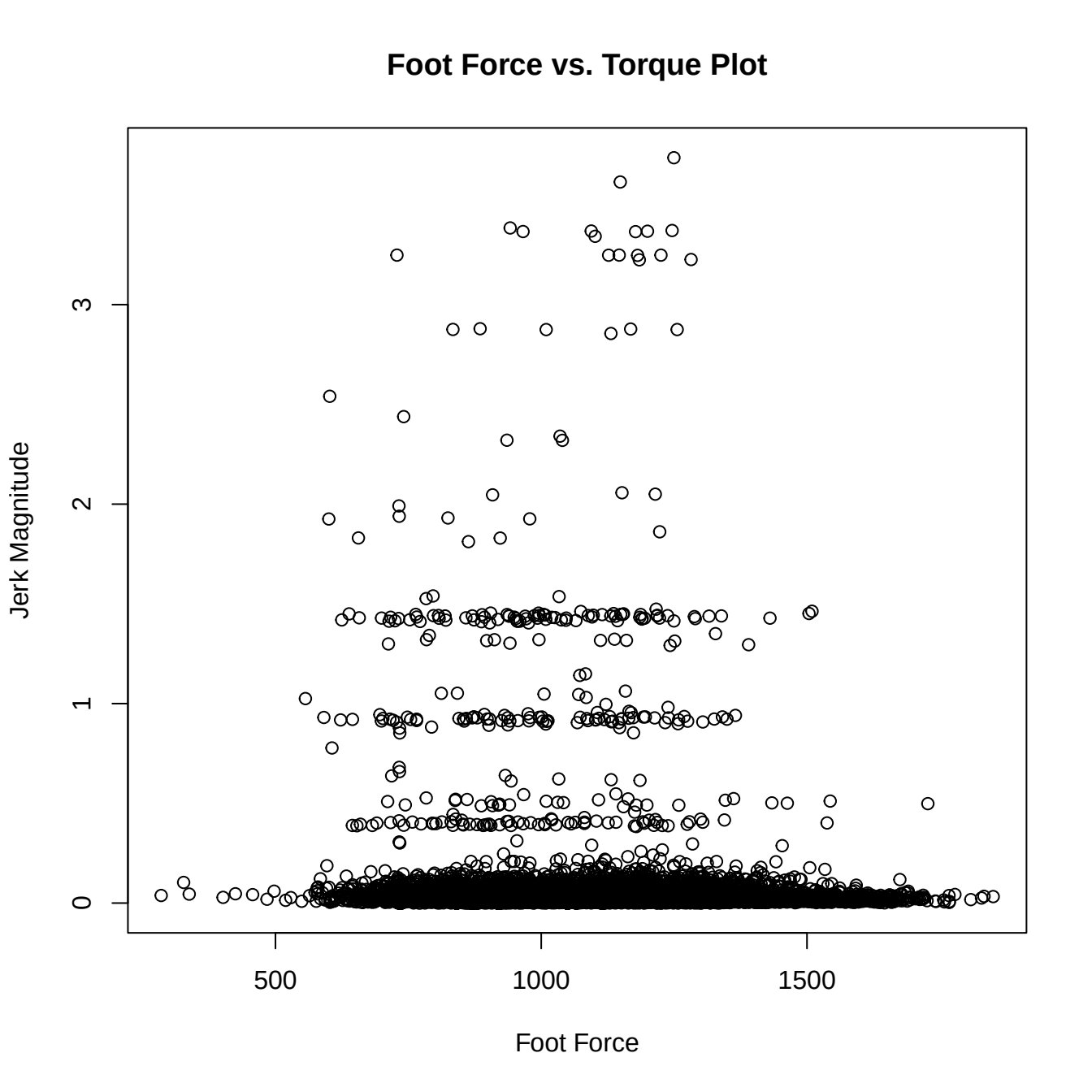

Histogram of Foot Force Derivative
Frequency
0
500
1000
1500
2000
−5000
0
5000
robot_data$d_footforce_dt

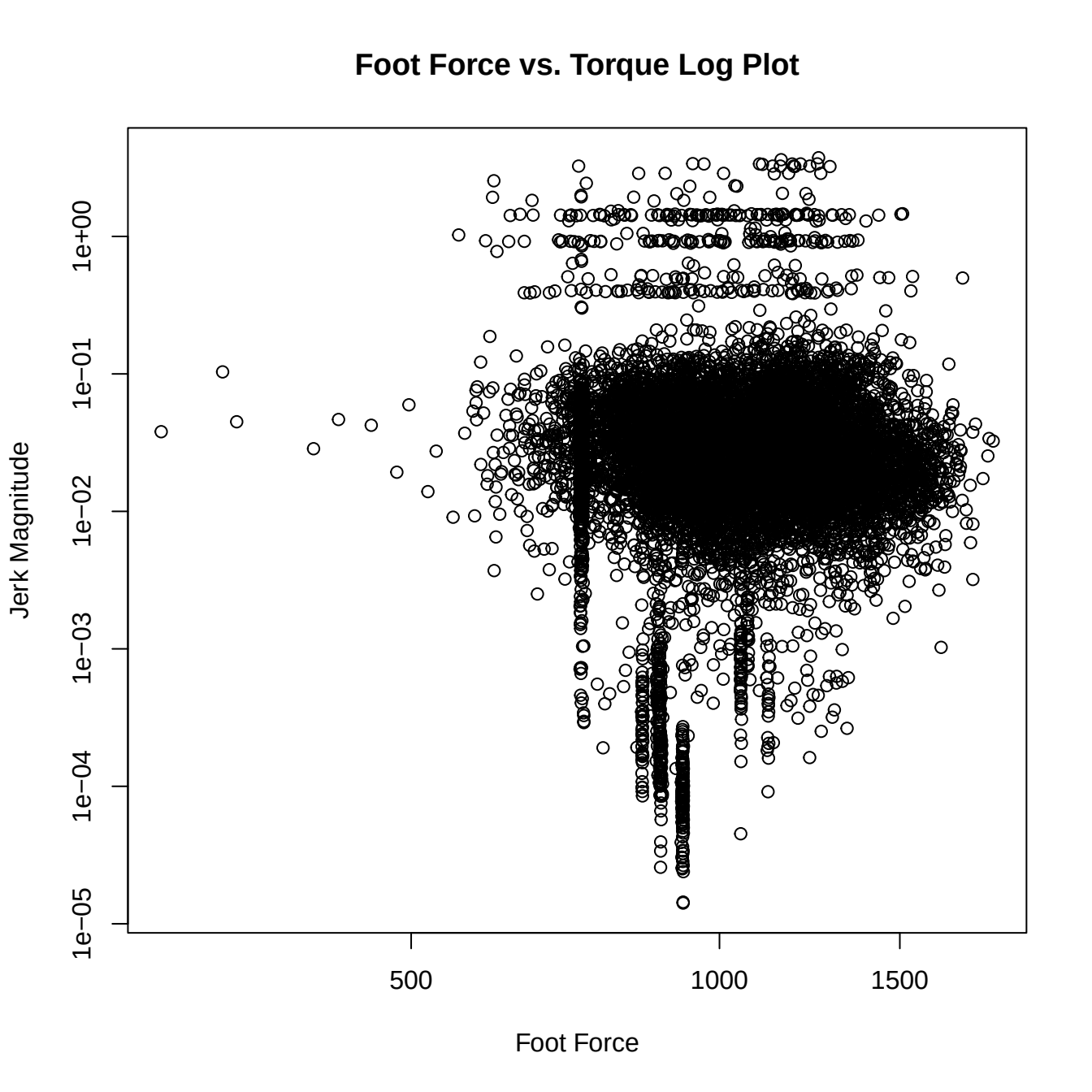

Histogram of Torque
Frequency
0
2000
4000
6000
8000
0
1
2
3
robot_data$total_torque

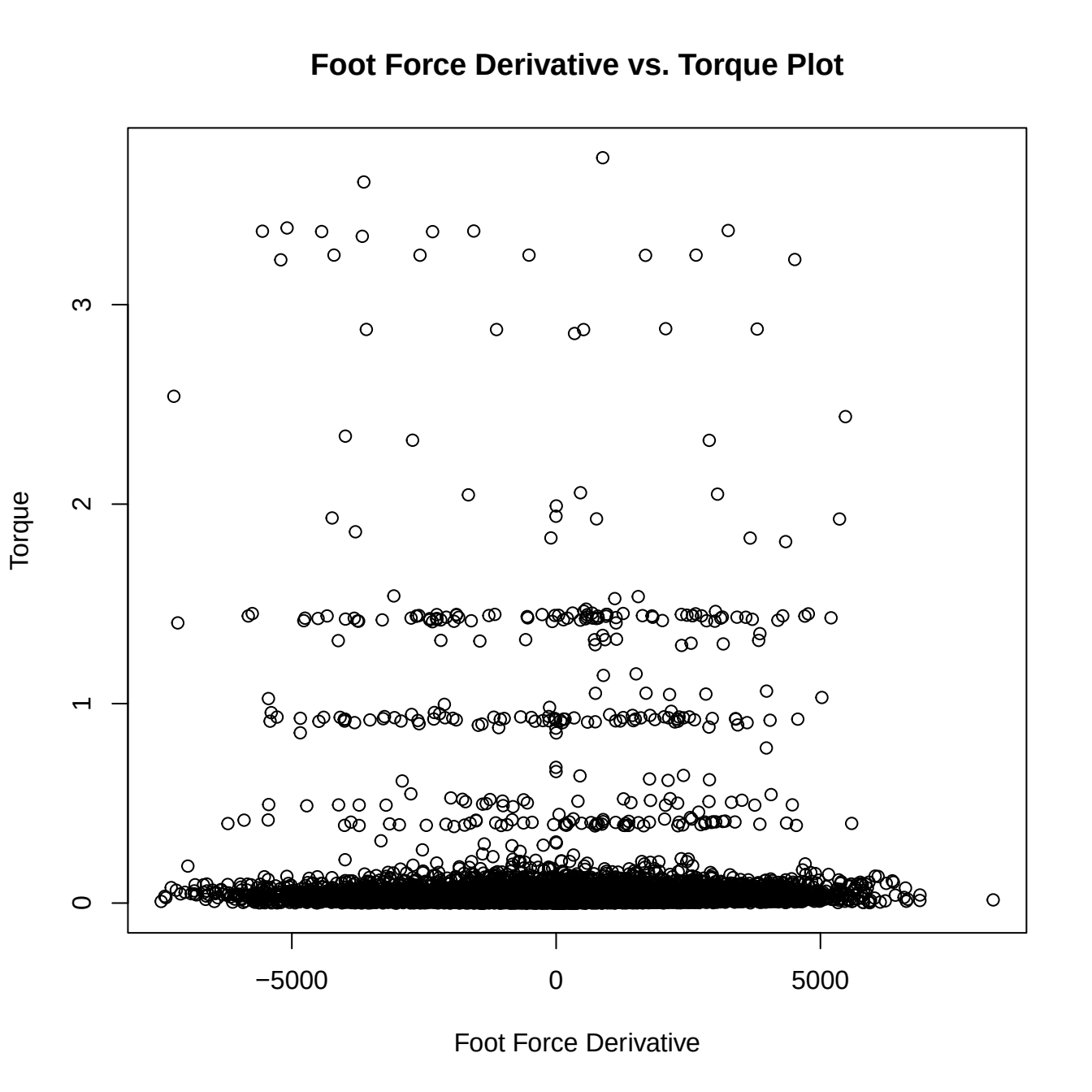

Histogram of Log−Transformed Foot Force
Frequency
0
500
1000
1500
6.0
6.5
7.0
7.5
log(robot_data$footforce)

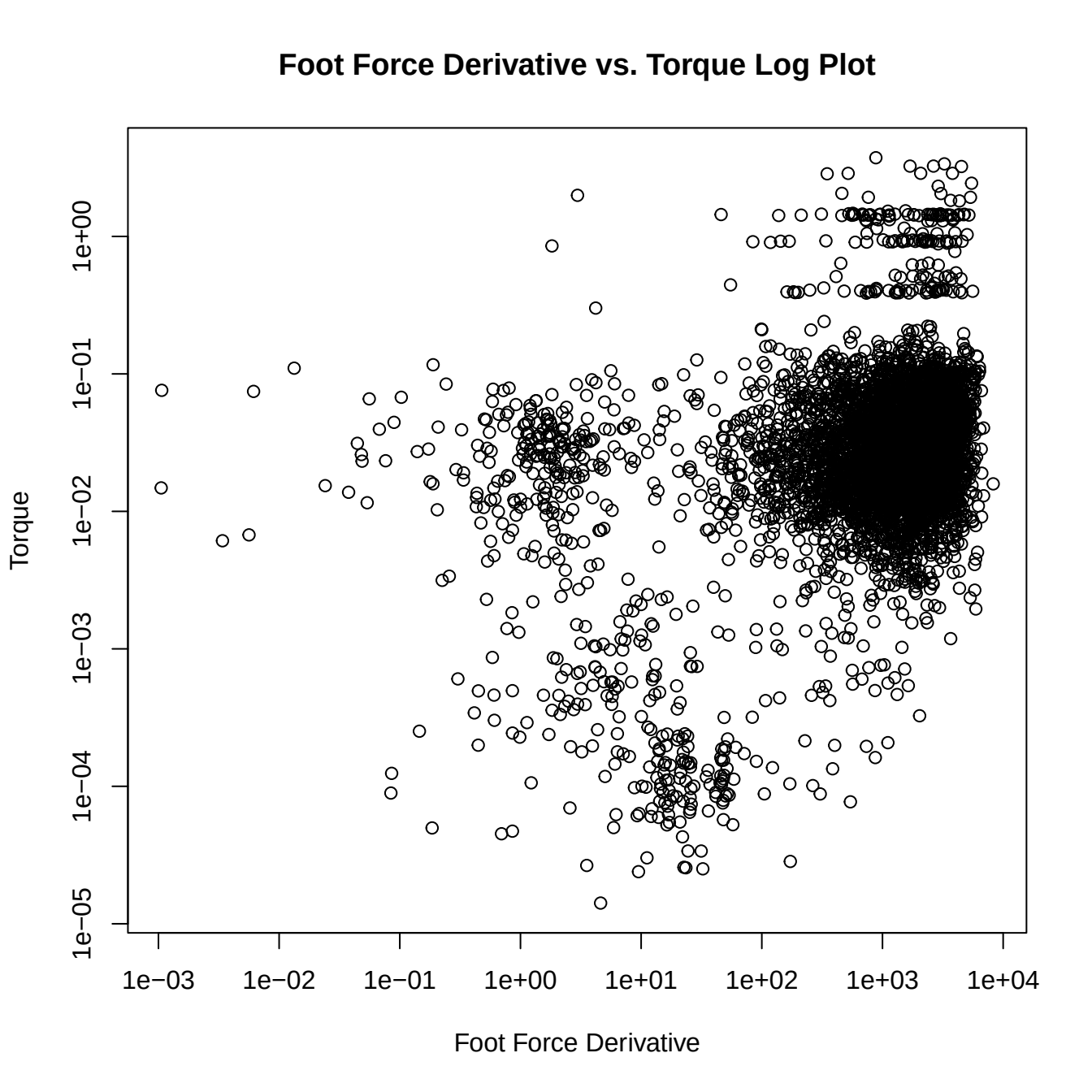

Log(|dFootForce/dt| + 1)
Frequency
0
500
1000
1500
0
2
4
6
8
robot_data$log_abs_dfootforce

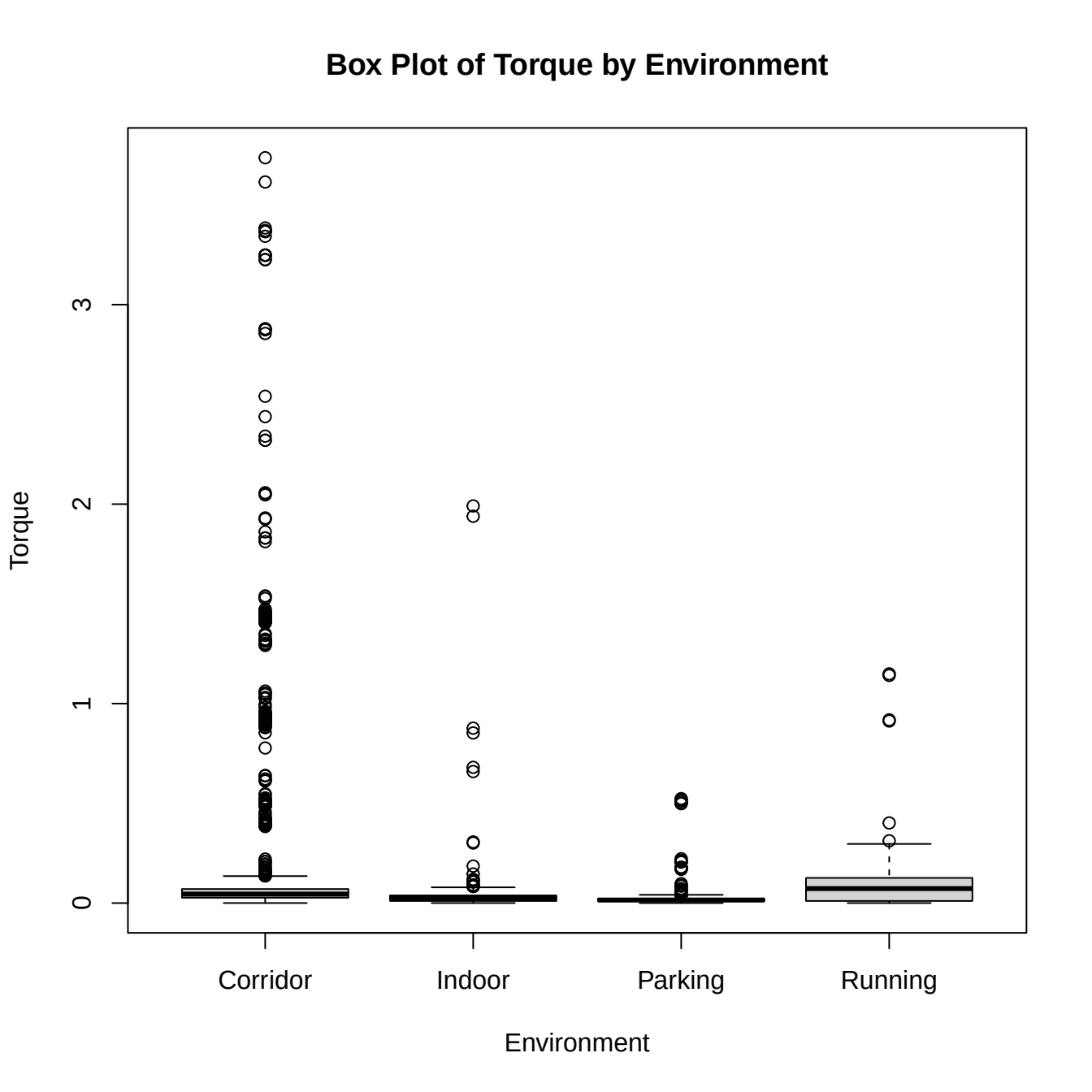

Histogram of Log−Transformed Torque
Frequency
0
500
1000
1500
2000
2500
3000
3500
−12
−10
−8
−6
−4
−2
0
2
log(robot_data$total_torque)

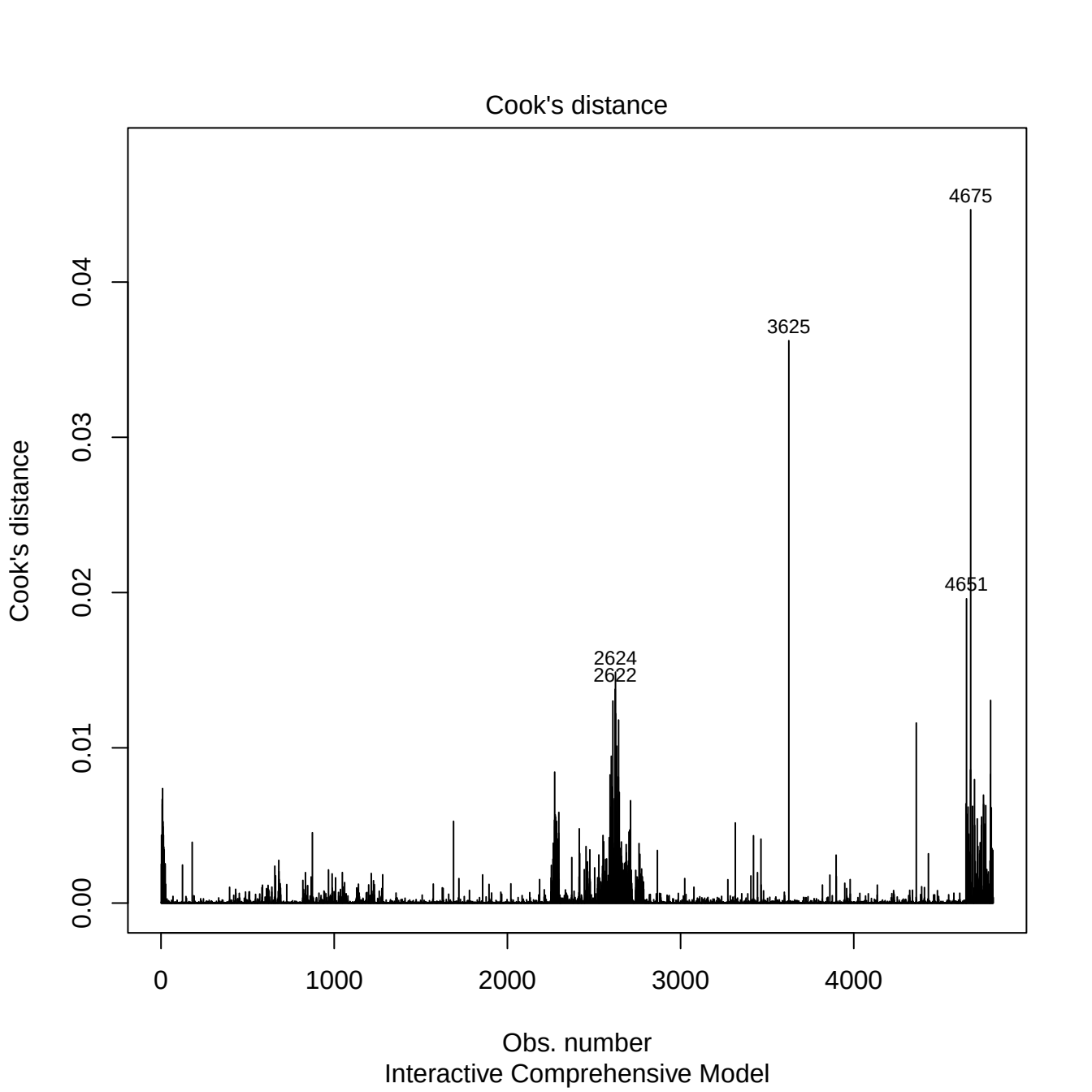

Foot Force vs. Torque Plot
Jerk Magnitude
0
1
2
3
500
1000
1500
Foot Force

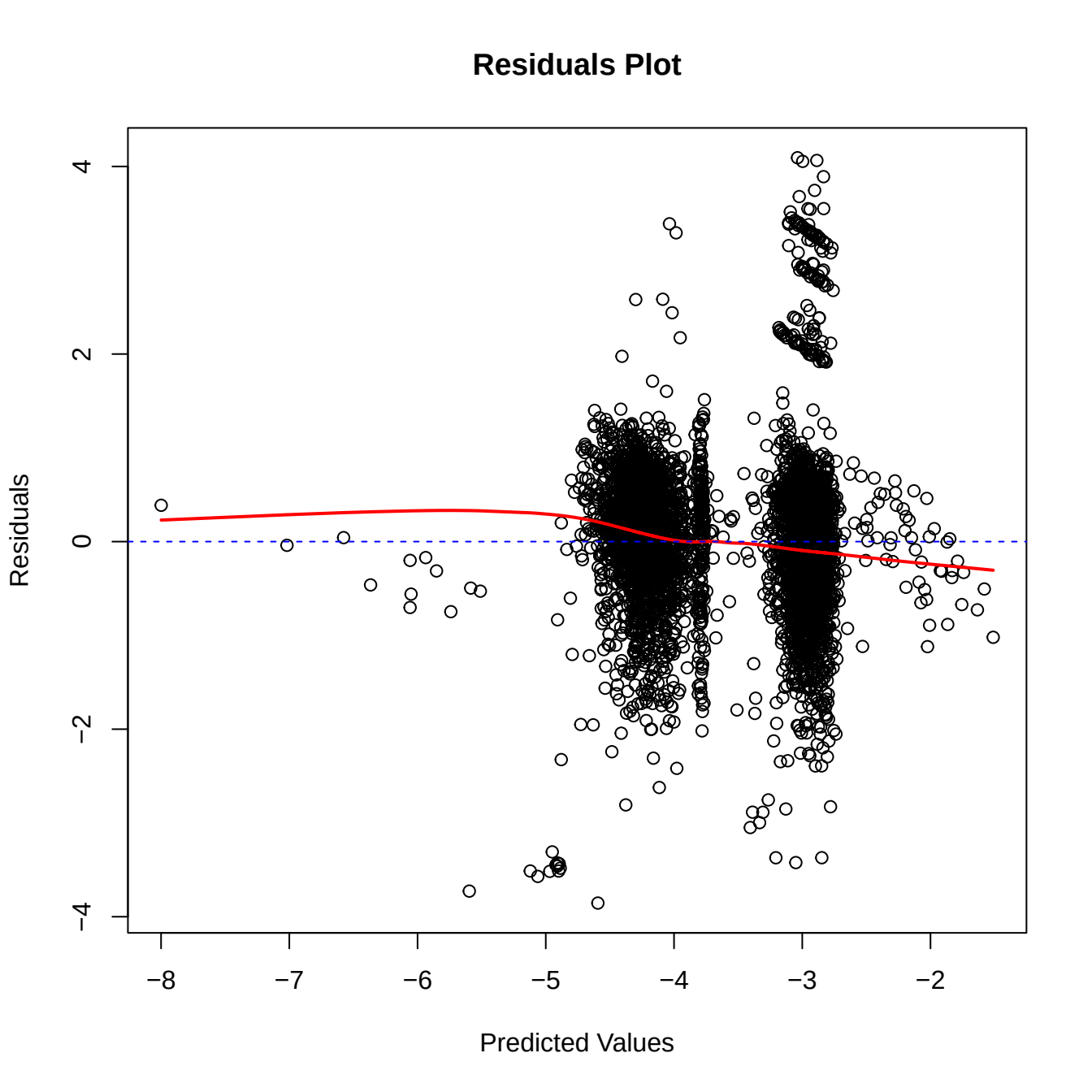

Foot Force vs. Torque Log Plot
Jerk Magnitude
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
500
1000
1500
Foot Force

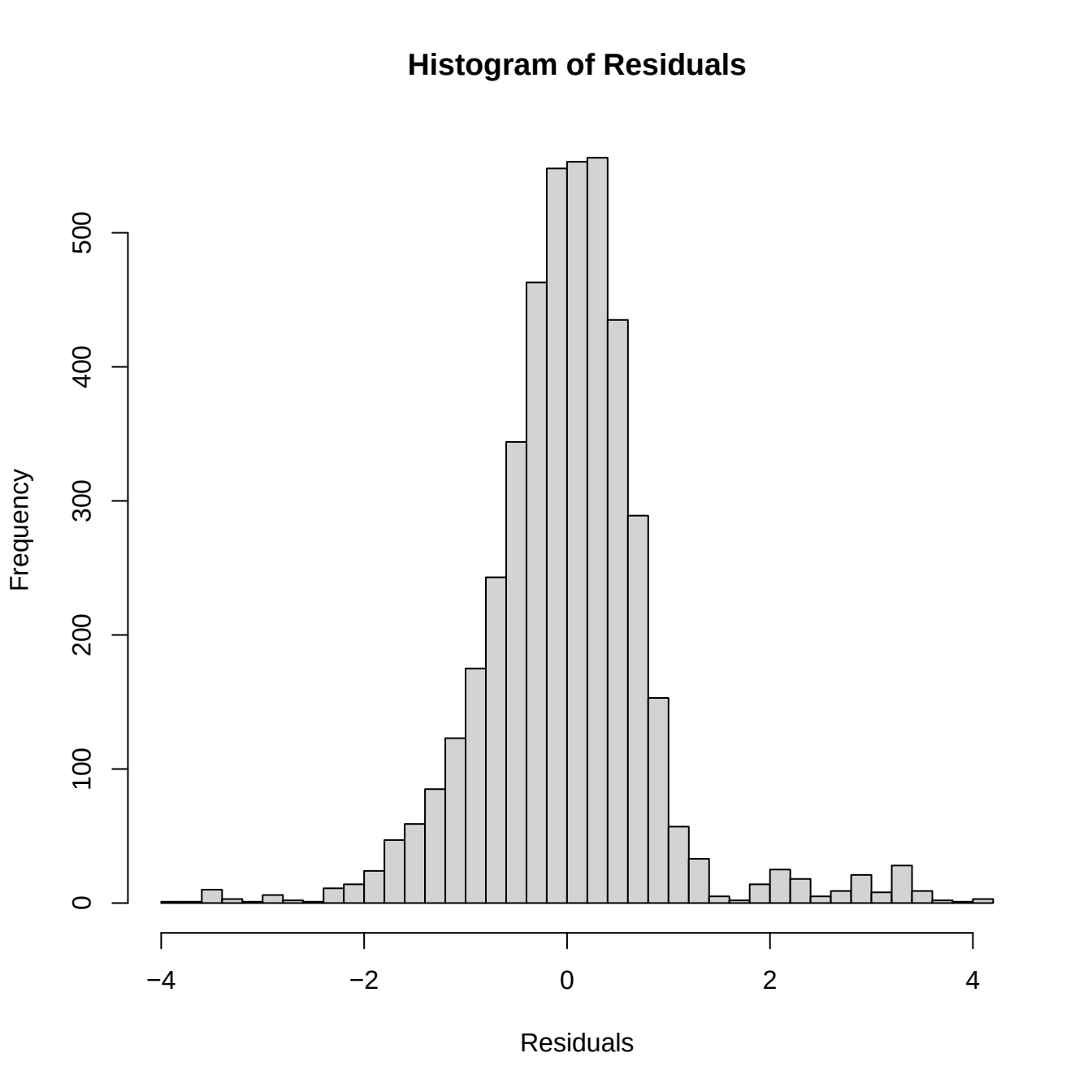

Foot Force Derivative vs. Torque Plot
Torque
0
1
2
3
−5000
0
5000
Foot Force Derivative

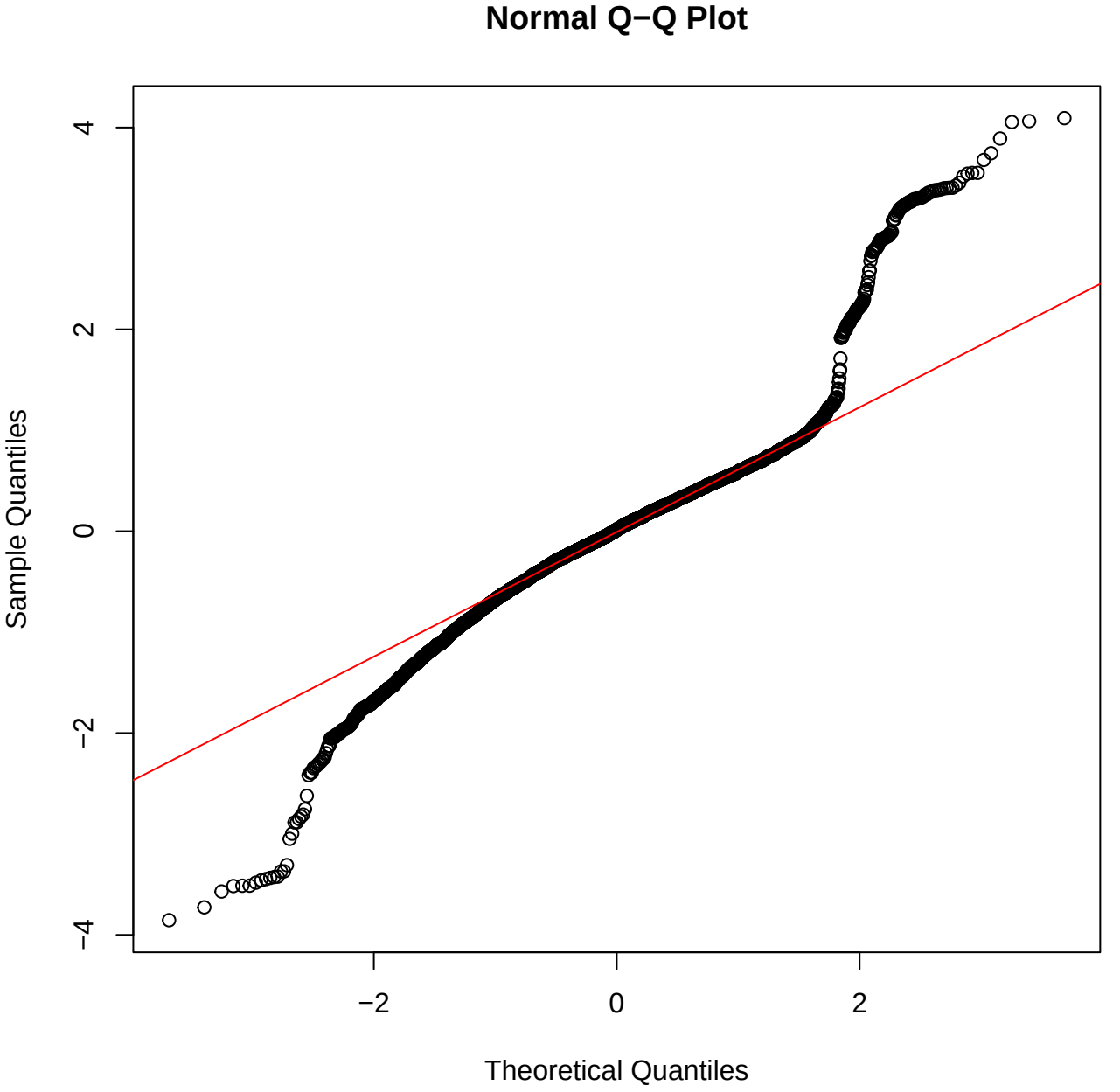

Foot Force Derivative vs. Torque Log Plot
Torque
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
1e+03
1e+04
Foot Force Derivative

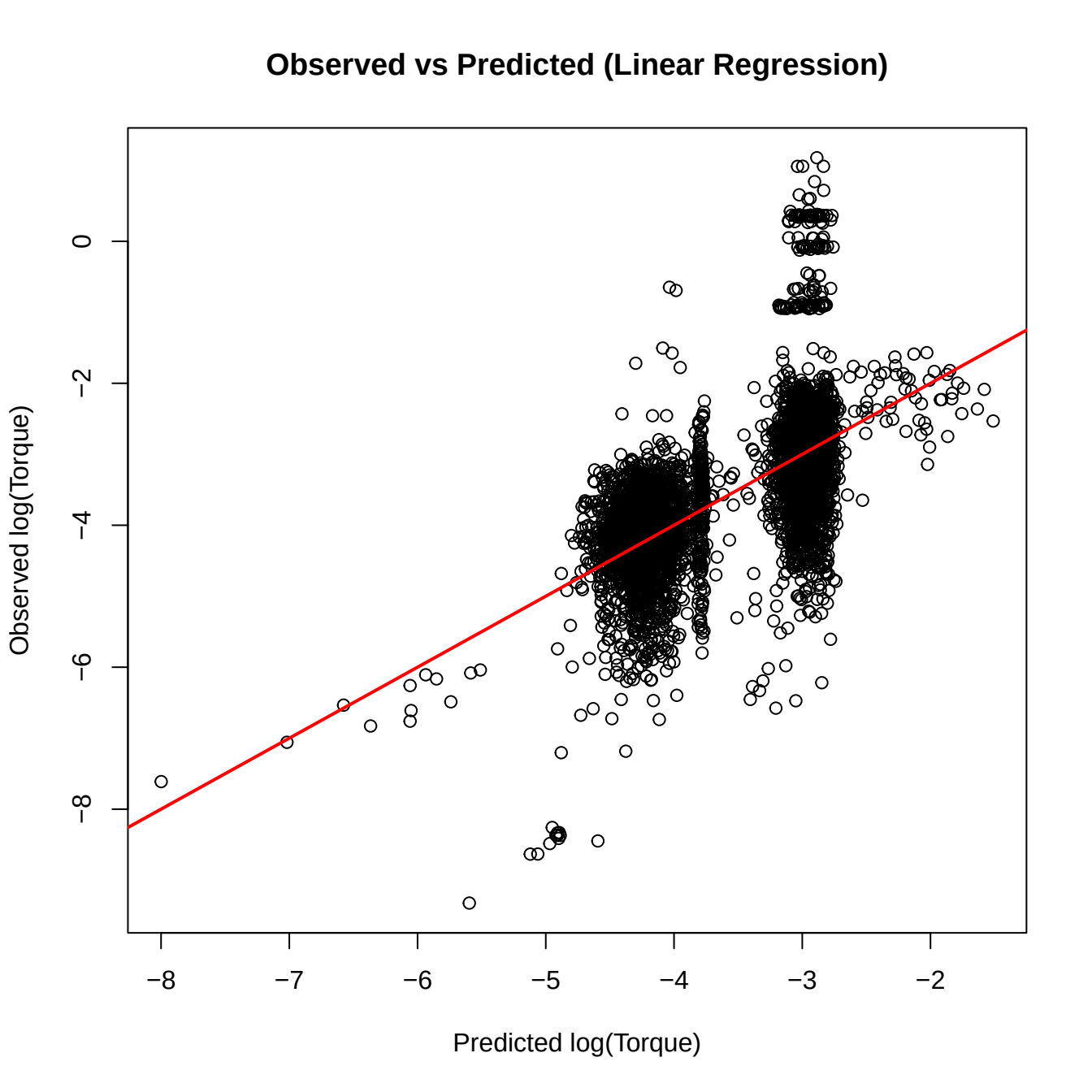

Box Plot of Torque by Environment
Torque
0
1
2
3
Corridor
Indoor
Parking
Running
Environment


Cook's distance
4675
3625
4651
2624
2622
0.00
0.01
0.02
0.03
0.04
Cook's distance
0
1000
2000
3000
4000
Obs. number
Interactive Comprehensive Model


Residuals Plot
Residuals
Predicted Values
4
2
0
−2
−4
−8
−7
−6
−5
−4
−3
−2


Histogram of Residuals
Frequency
0
100
200
300
400
500
−4
−2
0
2
4
Residuals


Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−4
−2
0
2


Observed vs Predicted (Linear Regression)
Observed log(Torque)
Predicted log(Torque)
0
−2
−4
−6
−8
−8
−7
−6
−5
−4
−3
−2